

Citations & Sources

Every numerical claim, physics statement, and corporate quote in "A Letter to Our Limited Partners," with its primary source. Verified against primary documents as of 9 June 2026. The fund and its author are fictional; none of the sources are.

FOOTNOTES 1 – 11

- 1 IBM Newsroom**, "IBM Debuts Next-Generation Quantum Processor & IBM Quantum System Two, Extends Roadmap to Advance Era of Quantum Utility," 4 December 2023. "IBM Condor, a 1,121 superconducting qubit quantum processor based on cross-resonance gate technology." Still the largest gate-based superconducting processor announced as of June 2026. newsroom.ibm.com
- 2 PsiQuantum / BusinessWire**, "PsiQuantum Raises \$1 Billion to Build Million-Qubit Scale, Fault-Tolerant Quantum Computers," 10 September 2025: "PsiQuantum was founded on the premise that commercially valuable quantum computing requires error correction—and therefore on the order of a million physical qubits." businesswire.com
- 3 Google Quantum AI & collaborators**, "Quantum error correction below the surface code threshold," *Nature* 638, 920–926 (2025; published online 9 December 2024). 101-qubit distance-7 surface-code logical qubit; "achieving a 10^{-6} error rate would require a distance-27 logical qubit using 1,457 physical qubits." nature.com/articles/s41586-024-08449-y (arXiv:2408.13687)
- 4 S. Bravyi, A. Cross, J. Gambetta, D. Maslov, P. Rall, T. Yoder (IBM)**, "High-threshold and low-overhead fault-tolerant quantum memory," *Nature* 627, 778–782 (27 March 2024). Circuit-level noise simulations: bivariate-bicycle codes preserve 12 logical qubits for $\sim 10^6$ syndrome cycles using 288 physical qubits, assuming 0.1% physical error rates. Not yet demonstrated on hardware as of June 2026. nature.com/articles/s41586-024-07107-7 (arXiv:2308.07915)
- 5 IBM Newsroom**, "IBM Unveils New Roadmap to Practical Quantum Computing Era; Plans to Deliver 4,000+ Qubit System," 10 May 2022: "All three of these scalability techniques will be leveraged toward IBM's 2025 goal: a 4,000+ qubit processor built with multiple clusters of modularly scaled processors." newsroom.ibm.com

- 6 **IBM Newsroom**, "IBM Sets the Course to Build World's First Large-Scale, Fault-Tolerant Quantum Computer at New IBM Quantum Data Center," 10 June 2025. Starling: "running 100 million quantum operations using 200 logical qubits," delivered by 2029; Blue Jay: "capable of executing 1 billion quantum operations over 2,000 logical qubits," roadmapped 2033. newsroom.ibm.com
- 7 **IonQ, Inc.**, Annual Report on Form 10-K, FY ended 31 December 2024 (filed 26 February 2025): FY2024 revenue \$43.1M (vs. \$22.0M in 2023). Customer concentration per 2024 Forms 10-Q: two customers = 69% (Q1), 75% (H1), 79% (nine months) of total revenue. AFRL award of \$54.5M announced 25 September 2024, the largest U.S. quantum contract of 2024. FY2025 revenue surpassed \$100M. SEC EDGAR (10-K) · ionq.com (AFRL)
- 8 **John Preskill**, "Beyond NISQ: The Megaquop Machine," *ACM Transactions on Quantum Computing* 6(3), Article 18, pp. 1–7 (29 April 2025); keynote, Q2B 2024 Silicon Valley, 11 December 2024. "There is no proposed application of NISQ computing with commercial value for which quantum advantage has been demonstrated when compared to the best classical hardware running the best algorithms for solving the same problems. Nor are there persuasive theoretical arguments indicating that commercially viable applications will be found that do not use quantum error-correcting codes and fault-tolerant quantum computing." DOI 10.1145/3723153 · preskill.caltech.edu
- 9 **Global Risk Institute** (M. Mosca & M. Piani), Quantum Threat Timeline Reports. 2024 edition (32 experts): probability of a cryptographically relevant quantum computer within 10 years, 19–34% across pessimistic-to-optimistic readings (within 5 years: 5–14%). 2025 edition (26 experts): within 10 years, 28–49%. 2024 report · 2025 report
- 10 **Quantum sensing**: Cerca Magnetics (University of Nottingham spin-out), commercial wearable OPM-MEG brain scanners: cercamagnetics.com; CSIRO LANDTEM SQUID sensors, two decades of commercial mineral-exploration use, credited with multi-billion-dollar ore discoveries: csiropedia.csiro.au; Q-CTRL Ironstone Opal quantum navigation, airborne defense trials and a 144-hour Royal Australian Navy sea trial (2025): q-ctrl.com
- 11 **Post-quantum cryptography migration**: NIST, "NIST Releases First 3 Finalized Post-Quantum Encryption Standards" (FIPS 203 ML-KEM, FIPS 204 ML-DSA, FIPS 205 SLH-DSA), 13 August 2024: nist.gov; NIST IR 8547 (initial public draft, November 2024), which proposes RSA-2048/ECC deprecated after 2030 and disallowed after 2035: csrc.nist.gov; NSA CNSA 2.0, requiring new national-security acquisitions to comply from 1 January 2027 with exclusive use by 2033: nsa.gov (FAQ)

EXHIBIT QUOTES

"Starling will be able to access the computational power required for these problems by running 100 million quantum operations using 200 logical qubits."

EXHIBIT B: IBM NEWSROOM, 10 JUNE 2025 (VERIFIED VERBATIM)

"No current quantum computers, including our quantum hardware, have reached a broad quantum advantage, and they may never reach such advantage."

EXHIBIT C: IONQ, INC., FORM 10-K RISK FACTORS; CARRIED IN EVERY ANNUAL FILING SINCE ITS 2021 LISTING (VERIFIED IN FY2022–FY2025 10-KS)

CALCULATOR MODEL: EXHIBIT A

years to advantage = $\log_2((\text{logical qubits required} \times \text{overhead}) \div 1,121) \times \text{doubling cadence}$, counted from 2026, rounded up. Anchor: IBM Condor, 1,121 physical qubits (footnote 1; substituting Atom Computing's 1,180-qubit neutral-atom record moves no estimate by more than a quarter-year). Logical-qubit requirements: chemistry ≈ 150 , optimization $\approx 1,000$, RSA-2048 $\approx 4,000$, per published resource estimates. Overhead slider: 100:1 floor = best demonstrated (footnote 3) to 1,500:1 ceiling = Google's projection for fault-tolerant error rates (footnote 3). Industry-comparison years (chemistry 2027, optimization 2029, RSA 2030) triangulate public vendor roadmaps and guidance. The model deliberately assumes the doubling cadence never slips and error correction works at scale on the first try.

Prepared by Jan Weichert, Caspar Miebach, and Bader Al Eisa for the Deep Tech Venturing & Investment module, IE Business School, June 2026. Aperture Quantum Partners and Maya Chen are fictional; every source above is real. All URLs accessed and verified 9 June 2026. Built with AI assistance (Anthropic's Claude); all citations human-reviewed against the primary documents.